Lecture 6 Exercise Solutions

1. Show that $x \equiv 2 \mod 6$ and $x \equiv 3 \mod 4$ have no simultaneous solutions.

   Assume there is an x that is a solution to both equations. Then

   $$x = 6k + 2 \Rightarrow x - 2 = 6k$$
   $$x = 4n + 3 \Rightarrow x - 3 = 4n$$

   Subtracting the second equation from the first we obtain

   $$1 = 6k - 4n \Rightarrow 6k = 4n + 1$$

   But, $6k$ is divisible by 2 and $4n+1$ is not. Therefore they cannot be equal and our assumption was wrong.

2. Show that $x \equiv 2 \mod 6$ and $x \equiv 0 \mod 4$ do have simultaneous solutions.

   $$x = 6k + 2 \Rightarrow x - 2 = 6k$$
   $$x = 4n + 0 \Rightarrow x = 4n$$

   Subtracting the second equation from the first we obtain

   $$-2 = 6k - 4n \Rightarrow 2n - 1 = 3k$$

   We can select $n = 5$ and $k = 3$ as one of the solutions—this is fine as it satisfies the equation. So $x = 4n = 4 \cdot 5 = 20$. This is, at least, one solution.

3. Find an integer x such that $x \equiv a \mod 12$ and $x \equiv 8 \mod 35$.

   Here we have two groups: $Z_{12}^{\times}$ and $Z_{35}^{\times}$. We can map these two groups into a composite group of modulus $12 \cdot 35 = 420$, that is $Z_{420}^{\times}$. Maybe we can find a solution by solving the problem there.

   Let $m = 12$, $n = 35$ and find and s and t, such that $s \cdot 12 + t \cdot 35 = 1$.

   We can use the Extended Euclidian Algorithm to do this, but it's quite clear that $s = 3$, $t = -1$, so $3 \cdot 12 + (-1)35 = 1$.

To find the value of x in Z_{420} we calculate

$$a(tn)+b(sm)=$$
$$a\cdot((-1)\cdot 35)+8\cdot(3\cdot 12)$$
$$a\cdot(-35)+288$$

Obviously, there are multiple solutions to this depending on the value of a:

$\overline{a\cdot(-35)+288}\in Z_{420}$ or $a\cdot(-35)+288 \bmod 420$

If $a=1$ then

$1\cdot(-35)+288=-35+288=\overline{253}\in Z_{420}$

4. Find an integer x such that $3x\equiv 2 \bmod 5$ and $4x\equiv 5 \bmod 7$.

Let's take this in multiple steps. But first, let's label the two moduli:

$m=5$
$n=7$

a) Find x such that $3x\equiv 2 \bmod 5$

$3x\equiv 2 \bmod 5 \Rightarrow 3x=5k+2$

So, we need to find a value of $k \ni 5k+2$ is divisible by 3. Obviously

$k=2\Rightarrow 5k+2=12=3x\Rightarrow x=4$. So, $\overline{4}\in Z_5$ and $a=4 \bmod m$.

b) Find x such that $4x\equiv 5 \bmod 7$

$4x\equiv 5 \bmod 7 \Rightarrow 4x=7k+5$

So, we need to find a value of $k \ni 7k+5$ is divisible by 4.

$k=1\Rightarrow 7k+5=12=4x\Rightarrow x=3$. So, $\overline{3}\in Z_7$ and $b=3 \bmod n$.

c) Determine s and t for $sm+tn=1$ where $m=5$ and $n=7$. We'll use the Extended Euclidean Algorithm.

gcd(7, 5) $7 = 1 \cdot 5 + 2$
gcd(5,2) $5 = 2 \cdot 2 + 1$

Reversing

$1 = 5 - 2 \cdot 2$
$1 = 5 - 2 \cdot (7 - 1 \cdot 5) = (-2) \cdot 7 + 3 \cdot 5 = -14 + 15$ Substituting 2

Thus $s = 3$ and $t = -2$

d) Calculate $x = a(tn) + b(sm)$

$s = 3 \wedge m = 5 \Rightarrow sm = 15$
$t = -2 \wedge n = 7 \Rightarrow tn = -14$

Therefore

$x = a(tn) + b(sm) = 4 \cdot (-14) + 3 \cdot 15 = -56 + 45 = -11$
But $-11 + 7 \cdot 5 = -11 + 35 = 24$, so $x = \overline{24} \in Z_{35}^{\times}$ and $\overline{24} \leftrightarrow (\overline{4}, \overline{3})$.

We can check this by the verifying

$24 \bmod 5 = 4 \Rightarrow \overline{4} \in Z_5$
$24 \bmod 7 = 3 \Rightarrow \overline{3} \in Z_7$

5. Explain why there are 8 different square roots of $1 \bmod 3 \cdot 5 \cdot 7 = 1 \bmod 105$.

   Clearly, according the Chinese Remainder Theorem, there is bijection such that $Z_{105}^x \Leftrightarrow Z_3^x \times Z_5^x \times Z_7^x$, and each of Z_3^x, Z_5^x, and Z_7^x has 2 square roots of $\overline{1}$, $\sqrt{\overline{1}} = \left\{ \overline{1}, \quad -\overline{1} \right\}$. So, for the bijection to be true Z_{105}^x must have $2 \cdot 2 \cdot 2 = 8$ square roots.

6. Is 2 a square modulo 101?

   $p = 101$ is prime. According to the Euler Criterion, if $\overline{y}^{(p-1)/2} = -\overline{1} \in Z_p^x$, then $\overline{y} \in Z_p^{\times}$ is NOT a square root. Therefore, we have to find $2^{(101-1)/2} \bmod 101$. By the fast exponentiation algorithm,

2	50	1
4	25	1
4	24	4
16	12	4
54	6	4
88	3	4
88	2	49
68	1	49
68	0	100

So $2^{50}\%101 = 100 \Rightarrow 2^{50} \equiv -1 \bmod 101$. Therefore, 2 is not a square modulo 101.

7. Find the principal square root of 2 modulo 19.

   $p = 19$ is prime, but to calculate a principle square root $p \equiv 3 \bmod 4$. Is it? Yes. $19 = 4 \cdot 4 + 3$. So, we can calculate the principle square root $\bar{x}^2 = \bar{2} \in Z_{19}^{\times}$, if it exists.

   $\bar{x} = \bar{2}^{(p+1)/4} = \bar{2}^{(19+1)/4} = \bar{2}^5 = \overline{13}$, so $\overline{13}$ is the principle square root of $\bar{2} \in Z_{19}^{\times}$, according to our calculations. Let's check it.

   $\overline{13}^2 = \overline{17} \neq \bar{2}$

   What's the problem? We forgot to check that $\bar{2}$ is a square in $Z_{19}^{\times}$.

   $2^{(19-1)/2} \bmod 19 = 2^9 \bmod 19 = 18 \equiv -1$

   So, $\bar{2}$ is NOT a square in $Z_{19}^{\times}$.

8. Find the principal square root of 6 modulo 19.

   Let's check that $\bar{6}$ is a square in $Z_{19}^{\times}$.

   $6^{(19-1)/2} \bmod 19 = 6^9 \bmod 19 = 1$, so we're good.

   $\bar{x} = \bar{6}^{(p+1)/4} = \bar{6}^{(19+1)/4} = \bar{6}^5 = \bar{5}$

   Let's check it.

$\overline{5}^2 = \overline{25} \equiv \overline{6} \in Z_{19}^{\times}$

9. Find the principal square root of 892 modulo 1039.

   First we check if $1039 \equiv 3 \bmod 4$ and it is. We'll just assume it's prime for the moment, because we haven't covered primality tests yet.

   Next, is $\overline{892}$ a square in $Z_{1039}^{\times}$?

   $892^{(1039-1)/2} \bmod 1039 = 892^9 \bmod 1039 = 1$. We're good.

   $\overline{x} = \overline{892}^{(p+1)/4} = \overline{892}^{(1039+1)/4} = \overline{892}^{260} = \overline{928}$

   Checking…

   $\overline{928}^2 = \overline{892} \in Z_{1039}^{\times}$

10. Find a cube root of 2 modulo 101. That is, find a cube root of $\overline{2}$ in $Z_{101}^{\times}$.

    101 is prime, therefore $Z_{101}^{\times} = 101 - 1 = 100$. Since 3 is prime and doesn't divide 100, we know that $\gcd(3,100) = 1$ and 3 has an inverse in $Z_{100}^{\times}$. So, we can apply Theorem 7. However, since 3 does not divide 100 we cannot apply Theorem 8 or Theorem 10.

    We can easily calculate the inverse using either the Extended Euclidean Algorithm or our Fast Exponentiation trick. To do the latter, we need to know $\left|Z_{100}^{\times}\right|$. This is easy if we know the factorization of $100 = 2^2 \cdot 5^2$, so

    $$\left|Z_{100}^{\times}\right| = (p_1 - 1)p_1^{e_1 - 1} \cdot (p_2 - 1)p_2^{e_2 - 1} = (2-1) \cdot 2^{2-1} \cdot (5-1) \cdot 5^{2-1} = 1 \cdot 2 \cdot 4 \cdot 5 = 40$$

    So, $\overline{3}^{-1} = \overline{3}^{40-1} = \overline{3}^{39} = \overline{67} \in Z_{100}^{\times}$. Therefore, $\overline{2}^{67} = \overline{26} \in Z_{101}^{\times}$ and $\overline{26}^3 = \overline{2} \in Z_{101}^{\times}$.

11. Find a cube root of 46 modulo 101. That is, find a cube root of $\overline{46}$ in $Z_{101}^{\times}$.

    Continuing from the previous problem, $\overline{3}^{-1} = \overline{3}^{40-1} = \overline{3}^{39} = \overline{67} \in Z_{100}^{\times}$. So, $\overline{46}^{67} = \overline{59} \in Z_{101}^{\times}$ and $\overline{59}^3 = \overline{46} \in Z_{101}^{\times}$.

12. Find a cube root of 61 modulo 103. That is, find a cube root of $\overline{61}$ in $Z_{103}^{\times}$.

103 is prime, therefore $Z_{103}^{\times} = 103 - 1 = 102$. But, since $\gcd(3,102) = 3$, meaning that they're not relatively prime, we cannot apply Theorem 7, because $3 \notin Z_{102}^{\times}$ and has no inverse in $Z_{102}^{\times}$. Fortunately, we can apply Theorems 8 and 10 because $3 \mid 102$. Let's do so.

Theorem 8 allows us to determine if there's a root for a particular element in a group.

$102 = 3 \cdot 34$

$\overline{61}^{34} = \overline{1} \in Z_{103}^{\times}$

So, yes 61 has a cube root in $Z_{103}^{\times}$.

Let's apply Theorem 10.

$102 = 3 \cdot 34$

Now, we calculate $\overline{3}^{-1} \in Z_{34}^{\times}$.

$34 = 11 \cdot 3 + 1 \Rightarrow (-11) \cdot 3 + 1 \cdot 34$

$\overline{3}^{-1} - \overline{11} \equiv \overline{23} \in Z_{34}^{\times}$

$\sqrt[3]{\overline{61}} = \overline{61}^{23} = \overline{34} \in Z_{103}^{\times}$

Let's check it.

$\overline{34}^{3} = \overline{61} \in Z_{103}^{\times}$

13. Solve $x^2 \equiv -1 \bmod 13 \cdot 17 \cdot 29$.

Based on the Sun Ze Theorem, the collection of integers modulo $13 \cdot 17 \cdot 29 = 6409$ satisfying $x^2 \equiv -1 \bmod 6409$ is a bijection with the set of triplets (x_1, x_2, x_3) such that $x_1^2 \equiv -1 \bmod 13$, $x_2^2 \equiv -1 \bmod 17$, and $x_3^2 \equiv -1 \bmod 29$, where $x_1 \in \mathrm{Z}_{13}, x_2 \in \mathrm{Z}_{17}, x_3 \in \mathrm{Z}_{29}$.

The bijection is $x \bmod 6409 \leftrightarrow (x \bmod 13, x \bmod 17, x \bmod 29)$. Thus once the right hand side of the bijection is known, the left hand side can be found. So,

we need to solve $x_1^2 \equiv -1 \bmod 13$, $x_2^2 \equiv -1 \bmod 17$, and $x_3^2 \equiv -1 \bmod 29$. Then we can map back to $x^2 \equiv -1 \bmod 6409$.

a) Solve $x_1^2 \equiv -1 \bmod 13$

Unfortunately, 13 is not congruent to $3 \bmod 4$. So, we can't use our method for finding the principle square root. But, we have two other methods.

First, Theorem 7 says that if $|G| = N \rightarrow \sqrt[n]{y} = y^r \in G \; where \; r = n^{-1} \in Z_N^\times$. In this case $n = 2, G = Z_{13}^\times, N = |Z_{13}^\times| = 12$. Now, we need to find $r = n^{-1} = 2^{-1} \in Z_{12}^\times$. Unfortunately, that's not possible either because $\gcd(2,12) \neq 1$. Obviously, this method will never work for the square root is $p-1$ is always even and can never be relatively prime to 2.

A second option is to apply Theorem 10 which states $\sqrt[n]{y} = y^r \in Z_p^\times \; where \; |Z_p^\times| = p - 1 = nk \; and \; r = n^{-1} \; and \; \gcd(n,k) = 1$. Unfortunately, $p - 1 = 13 - 1 = 12 = 2 \cdot 6 = n \cdot k$, but $\gcd(n,k) = \gcd(2,6) \neq 1$. So, we can't use that theorem either.

Fortunately, $Z_{13}^\times$ isn't that big. We want to find a $1 < x_1 < 13$, where when we square it is congruent to $-\bar{1}$: $-\bar{1} \equiv \overline{12} \equiv \overline{25} \in Z_{13}^\times$. Clearly, $\overline{25}$ is a square and $x_1 = \sqrt{\overline{25}} = \left\{ \; -\bar{5}, \quad \bar{5} \; \right\} \in Z_{13}^\times$.

b) Solve $x_2^2 \equiv -1 \bmod 17$

Again, 17 is not congruent to $3 \bmod 4$. So, we can't use our method for finding the principle square root.

Proceeding, as before, $-\bar{1} \equiv \overline{16} \in Z_{17}^\times$. So, $x_2 = \sqrt{\overline{16}} = \left\{ \; -\bar{4}, \quad \bar{4} \; \right\} \in Z_{17}^\times$

c) Solve $x_3^2 \equiv -1 \bmod 29$

Finally, 29 is not congruent to $3 \bmod 4$. So, we can't use our method for finding the principle square root.

Proceeding, as before, $-\bar{1} \equiv \overline{28} \equiv \overline{57} \equiv \overline{86} \equiv \overline{115} \equiv \overline{144} \in Z_{29}^\times$. So, $x_3 = \sqrt{\overline{144}} = \left\{ \; -\overline{12} \quad \overline{12} \; \right\} \in Z_{29}^\times$.

d) Enumerate the pre-images. Obviously there are $2^3 = 8$ such pre-images

i. $(x_1 \equiv 5 \bmod 13,\ x_2 \equiv 4 \bmod 17,\ x_3 \equiv 12 \bmod 29)$
ii. $(x_1 \equiv 5 \bmod 13,\ x_2 \equiv 4 \bmod 17,\ x_3 \equiv -12 \bmod 29)$
iii. $(x_1 \equiv 5 \bmod 13,\ x_2 \equiv -4 \bmod 17,\ x_3 \equiv 12 \bmod 29)$
iv. $(x_1 \equiv 5 \bmod 13,\ x_2 \equiv -4 \bmod 17,\ x_3 \equiv -12 \bmod 29)$
v. $(x_1 \equiv -5 \bmod 13,\ x_2 \equiv 4 \bmod 17,\ x_3 \equiv 12 \bmod 29)$
vi. $(x_1 \equiv -5 \bmod 13,\ x_2 \equiv 4 \bmod 17,\ x_3 \equiv -12 \bmod 29)$
vii. $(x_1 \equiv -5 \bmod 13,\ x_2 \equiv -4 \bmod 17,\ x_3 \equiv 12 \bmod 29)$
viii. $(x_1 \equiv -5 \bmod 13,\ x_2 \equiv -4 \bmod 17,\ x_3 \equiv -12 \bmod 29)$

e) Calculate $x = (T_1 M_1 x_1 + T_2 M_2 x_2 + T_3 M_3 x_3) \bmod M$

$$M = m_1 \cdot m_2 \cdot m_3 = 13 \cdot 17 \cdot 29$$

$$M_1 = \frac{M}{m_1} = 17 \cdot 29 = 493 \equiv 12 \bmod 13$$

$$M_2 = \frac{M}{m_2} = 13 \cdot 29 = 377 \equiv 3 \bmod 17$$

$$M_3 = \frac{M}{m_3} = 13 \cdot 17 = 221 \equiv 18 \bmod 29$$

$$T_1 = M_1^{-1} \in Z_{13}^{\times}$$
$$\gcd(13,12) \rightarrow 13 = 1 \cdot 12 + 1$$

Reversing

$$1 = 1 \cdot 13 + (-1) \cdot 12 \Rightarrow T_1 = -1$$

$$\boxed{T_1 M_1 = (-1) \cdot 493 = -493}$$

$$T_2 = M_2^{-1} \in Z_{17}^{\times}$$
$$\gcd(17,3) \rightarrow 17 = 5 \cdot 3 + 2$$
$$\gcd(3,2) \rightarrow 3 = 1 \cdot 2 + 1$$

Reversing

$$1 = 1 \cdot 3 + (-1) \cdot 2$$
$$1 = 1 \cdot 3 + (-1) \cdot (17 + (-5) \cdot 3) = 6 \cdot 3 + (-1) \cdot 17 \Rightarrow T_2 = 6$$

$\boxed{T_2 M_2 = 6 \cdot 377 = 2262}$

$T_3 = M_3^{-1} \in Z_{29}^{\times}$

$\gcd(29,18) \rightarrow 29 = 1 \cdot 18 + 11$

$\gcd(18,11) \rightarrow 18 = 1 \cdot 11 + 7$

$\gcd(11,7) \rightarrow 11 = 1 \cdot 7 + 4$

$\gcd(7,4) \rightarrow 7 = 1 \cdot 4 + 3$

$\gcd(4,3) \rightarrow 4 = 1 \cdot 3 + 1$

Reversing

$1 = 1 \cdot 4 + (-1) \cdot 3$

$1 = 1 \cdot 4 + (-1) \cdot (7 + (-1) \cdot 4) = (-1) \cdot 7 + 2 \cdot 4$

$1 = (-1) \cdot 7 + 2 \cdot (11 + (-1) \cdot 7) = 2 \cdot 11 + (-3) \cdot 7$

$1 = 2 \cdot 11 + (-3) \cdot (18 + (-1) \cdot 11) = (-3) \cdot 18 + 5 \cdot 11$

$1 = (-3) \cdot 18 + 5 \cdot (29 + (-1) \cdot 18) = 6 \cdot 29 + (-8) \cdot 18 \Rightarrow T_3 = -8$

$\boxed{T_3 M_3 = (-8) \cdot 221 = -1768}$

Substituting

$\boxed{\boxed{x = (-493 \cdot x_1 + 2262 \cdot x_2 + (-1768) \cdot x_3) \bmod 6409}}$

f) Calculate the image of each of the pre-images

i. $(x_1 \equiv 5 \bmod 13,\ x_2 \equiv 4 \bmod 17,\ x_3 \equiv 12 \bmod 29)$

$x = (-493 \cdot x_1 + 2262 \cdot x_2 + (-1768) \cdot x_3) \bmod 6409$

$x = (-493 \cdot 5 + 2262 \cdot 4 + (-1768) \cdot 12) \bmod 6409 = -14633 \bmod 6409 = 4594$

Verify:

$4594^2 = 6408 \equiv -1 \bmod 6409$

ii. $(x_1 \equiv 5 \bmod 13,\ x_2 \equiv 4 \bmod 17,\ x_3 \equiv -12 \bmod 29)$

$x = (-493 \cdot x_1 + 2262 \cdot x_2 + (-1768) \cdot x_3) \bmod 6409$

$x = (-493 \cdot 5 + 2262 \cdot 4 + (-1768) \cdot (-12)) \bmod 6409 = 27799 \bmod 6409 = 2163$

Verify:

$2163^2 = 6408 \equiv -1 \bmod 6409$

iii. $(x_1 \equiv 5 \bmod 13,\ x_2 \equiv -4 \bmod 17,\ x_3 \equiv 12 \bmod 29)$

$x = (-493 \cdot x_1 + 2262 \cdot x_2 + (-1768) \cdot x_3) \bmod 6409$

$x = (-493 \cdot 5 + 2262 \cdot (-4) + (-1768) \cdot 12) \bmod 6409 = -32729 \bmod 6409 = 5725$

Verify:

$5725^2 = 6408 \equiv -1 \bmod 6409$

iv. $(x_1 \equiv 5 \bmod 13,\ x_2 \equiv -4 \bmod 17,\ x_3 \equiv -12 \bmod 29)$

$x = (-493 \cdot x_1 + 2262 \cdot x_2 + (-1768) \cdot x_3) \bmod 6409$

$x = (-493 \cdot 5 + 2262 \cdot (-4) + (-1768) \cdot (-12)) \bmod 6409 = 9703 \bmod 6409 = 3294$

Verify:

$3924^2 = 6408 \equiv -1 \bmod 6409$

v. $(x_1 \equiv -5 \bmod 13,\ x_2 \equiv 4 \bmod 17,\ x_3 \equiv 12 \bmod 29)$

$x = (-493 \cdot x_1 + 2262 \cdot x_2 + (-1768) \cdot x_3) \bmod 6409$

$x = (-493 \cdot (-5) + 2262 \cdot 4 + (-1768) \cdot 12) \bmod 6409 = -9703 \bmod 6409 = 3115$

Verify:

$3115^2 = 6408 \equiv -1 \bmod 6409$

vi. $(x_1 \equiv -5 \bmod 13,\ x_2 \equiv 4 \bmod 17,\ x_3 \equiv -12 \bmod 29)$

$x = (-493 \cdot x_1 + 2262 \cdot x_2 + (-1768) \cdot x_3) \bmod 6409$

$x = (-493 \cdot (-5) + 2262 \cdot 4 + (-1768) \cdot (-12)) \bmod 6409 = 32729 \bmod 6409 = 684$

$684^2 = 6408 \equiv -1 \bmod 6409$

vii. $(x_1 \equiv -5 \bmod 13,\ x_2 \equiv -4 \bmod 17,\ x_3 \equiv 12 \bmod 29)$

$x = (-493 \cdot x_1 + 2262 \cdot x_2 + (-1768) \cdot x_3) \bmod 6409$

$x = (-493 \cdot (-5) + 2262 \cdot (-4) + (-1768) \cdot 12) \bmod 6409 = -27799 \bmod 6409 = 4246$

Verify:

$4246^2 = 6408 \equiv -1 \bmod 6409$

viii. $(x_1 \equiv -5 \bmod 13,\ x_2 \equiv -4 \bmod 17,\ x_3 \equiv -12 \bmod 29)$

$x = (-493 \cdot x_1 + 2262 \cdot x_2 + (-1768) \cdot x_3) \bmod 6409$

$x = (-493 \cdot (-5) + 2262 \cdot (-4) + (-1768) \cdot (-12)) \bmod 6409 = 14633 \bmod 6409 = 1815$

Verify:

$1815^2 = 6408 \equiv -1 \bmod 6409$

Thus there 8 are solutions in $Z_{6409}^{\times}$ for the equation $x^2 = -\bar{1}$.

$$\bar{x} = \left\{ \overline{4594},\ \overline{2163},\ \overline{5725},\ \overline{3924},\ \overline{3115},\ \overline{684},\ \overline{4246},\ \overline{1815} \right\}$$

But in $Z_{6409}^{\times}$

$$\overline{4594} + \overline{1815} = \bar{0} \Rightarrow \overline{1815} = -\overline{4594}$$
$$\overline{2163} + \overline{4246} = \bar{0} \Rightarrow \overline{4246} = -\overline{2163}$$
$$\overline{5725} + \overline{684} = \bar{0} \Rightarrow \overline{684} = -\overline{5725}$$
$$\overline{3924} + \overline{3115} = \bar{0} \Rightarrow \overline{3115} = -\overline{3924}$$

So

$$\bar{x} = \left\{ \pm\overline{4594},\ \pm\overline{2163},\ \pm\overline{5725},\ \pm\overline{3924} \right\}$$

14. Find an integer x such that $x \equiv 2 \bmod 3$, $x \equiv 3 \bmod 5$, and $x \equiv 4 \bmod 7$.

Obviously, 3, 5, and 7 are mutually prime.

$$M = m_1 \cdot m_2 \cdot m_3 = 3 \cdot 5 \cdot 7$$
$$M_1 = \frac{M}{m_1} = m_2 \cdot m_3 = 5 \cdot 7 = 35$$
$$M_2 = \frac{M}{m_2} = m_1 \cdot m_3 = 3 \cdot 7 = 21'$$
$$M_3 = \frac{M}{m_3} = m_1 \cdot m_2 = 3 \cdot 5 = 15$$

$$T_1 = M_1^{-1} \bmod m_1$$
$$35 \equiv 2 \bmod 3$$
$$T_1 = 2^{-1} = 2 \bmod 3$$
$$\boxed{T_1 M_1 = 2 \cdot 35 = 70}$$

$T_2 = M_2^{-1} \bmod m_2$

$21 \equiv 1 \bmod 5$

$T_2 = 1^{-1} = 1 \bmod 5$

$\boxed{T_2 M_2 = 1 \cdot 21 = 21}$

$T_3 = M_3^{-1} \bmod m_3$

$15 \equiv 1 \bmod 7$

$T_3 = 1^{-1} = 1 \bmod 5$

$\boxed{T_3 M_3 = 1 \cdot 15 = 15}$

$x = (T_1 M_1 \cdot 2 + T_2 M_2 \cdot 3 + T_3 M_3 \cdot 4) \bmod 3 \cdot 5 \cdot 7$

$x = (T_1 M_1 \cdot 2 + T_2 M_2 \cdot 3 + T_3 M_3 \cdot 4) \bmod 105$

$x = (70 \cdot 2 + 21 \cdot 3 + 15 \cdot 4) \bmod 105$

$x = 53 \bmod 105$

Note that while solving the system of equations

$$\begin{cases} x \equiv b_1 \bmod m_1 \\ . \\ . \\ x \equiv b_n \bmod m_n \end{cases}$$

that most of the effort is about finding $T_i M_i$ for $i = 1 \cdots n$, because these products depend exclusively on $m_1 \cdots m_n$. Only at the last moment will $b_1 \cdots b_n$ be placed in the formula $x = (T_1 M_1 b_1 + \cdots + T_n M_n b_n) \bmod m_1 \cdots m_n$. So, if we have to solve a different system of equations, where $m_1 \cdots m_n$ are the same, but $b_1 \cdots b_n$ are different, it won't take any time for all the preliminary work will have been done.